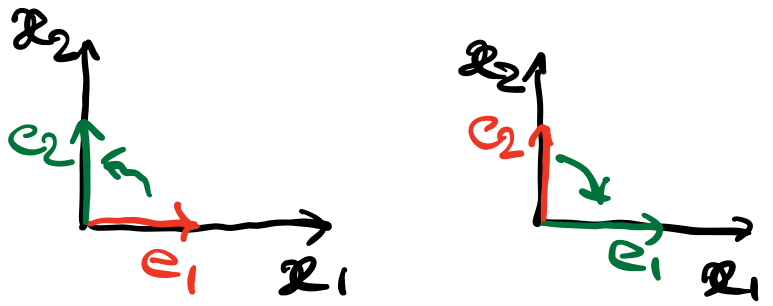
Lecture 5: Eigenvalues & Eigenvectors (5.1 & 5.2)

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$



Consider vector $\vec{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$A\vec{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \vec{x} = 1 \cdot \vec{x}$$

eigenvector, eigenvalue

Example

$$A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$$

$A\vec{x} = 7\vec{x}, \quad \vec{x} \neq 0$

$A\vec{x} - 7\vec{x} = 0$

$(A - 7I)\vec{x} = 0$

$$A - 7I = \begin{bmatrix} -6 & 6 \\ 5 & -5 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} -6 & 6 & 0 \\ 5 & -5 & 0 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$x_1 - x_2 = 0 \Rightarrow x_1 = x_2$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot x_2 \qquad \vec{x} \in \text{Span}\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$$

eigenvector for eigenvalue 7

Not unique! $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$ is also a solution.

Definition: A $n \times n$-matrix.

Eigenvalue $\lambda \in \mathbb{R}$: $A\vec{x} = \lambda\vec{x}$ for some $\vec{x} \neq 0$

↑ eigenvector associated with eigenvalue λ.

$A - \lambda I$ is singular (not invertible)
$\Rightarrow \det(A - \lambda I) = 0$

Important Message #1	Eigenvalue 'solves' the characteristic polynomial

Theorem: A $n \times n$ matrix. The following are equivalent:

- λ is an eigenvalue of A
- $A - \lambda I$ is singular (not invertible)
- $\det(A - \lambda I) = 0$

Example continued

$$A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$$

$\det(A - \lambda I) = 0$

$$\begin{vmatrix} 1-\lambda & 6 \\ 5 & 2-\lambda \end{vmatrix} = 0 \Rightarrow$$

characteristic polynomial

$$(1-\lambda)(2-\lambda) - 30 = 0$$
$$2 - \lambda - 2\lambda + \lambda^2 - 30 = 0$$
$$\lambda^2 - 3\lambda - 28 = 0$$
$$(\lambda - 7)(\lambda + 4) = 0$$
$$\lambda_1 = 7, \quad \lambda_2 = -4$$

eigenvalues

$\lambda_2 = -4$ Find eigenvectors

$(A - (-4)I)\vec{x} = \vec{0}$

$$A + 4I = \begin{bmatrix} 5 & 6 \\ 5 & 6 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 5 & 6 & 0 \\ 5 & 6 & 0 \end{array}\right] \sim \left[\begin{array}{cc|c} 5 & 6 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$$5x_1 + 6x_2 = 0$$
$$\Rightarrow x_1 = -\frac{6}{5}x_2$$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -\frac{6}{5} \\ 1 \end{bmatrix} \cdot x_2$$

$$\vec{x} \in \text{Span}\left\{\begin{bmatrix} -\frac{6}{5} \\ 1 \end{bmatrix}\right\}$$
$$= \text{Span}\left\{\begin{bmatrix} -6 \\ 5 \end{bmatrix}\right\}$$

$\lambda_1 = 7 \qquad \vec{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\lambda_2 = -4 \qquad \vec{v}_2 = \begin{bmatrix} -6 \\ 5 \end{bmatrix}$

$\{\vec{v}_1, \vec{v}_2\}$ is a linearly independent set of vectors (basis of $\mathbb{R}^2$).

$$P = \begin{bmatrix} \vec{v}_1 & \vec{v}_2 \end{bmatrix} = \begin{bmatrix} 1 & -6 \\ 1 & 5 \end{bmatrix}$$

A is diagonalizable

$$A \cdot P = \begin{bmatrix} A\vec{v}_1 & A\vec{v}_2 \end{bmatrix} = \begin{bmatrix} \lambda_1\vec{v}_1 & \lambda_2\vec{v}_2 \end{bmatrix} = P \cdot D$$

where $D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$

In other words $P^{-1}AP = D$

Properties: A $n \times n$, $\lambda_1, \lambda_2, \ldots, \lambda_p$ and $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p$

(1) $\{\vec{v}_1, \ldots, \vec{v}_p\}$ are linearly independent (because eigenvalues are distinct)

(2) If $p = n$: basis of eigenvectors

(3) If $p = n$ $P^{-1}AP = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_p \end{bmatrix}$ with $P = [\vec{v}_1 \ \vec{v}_2 \ \ldots \ \vec{v}_p]$

↑ diagonalizable

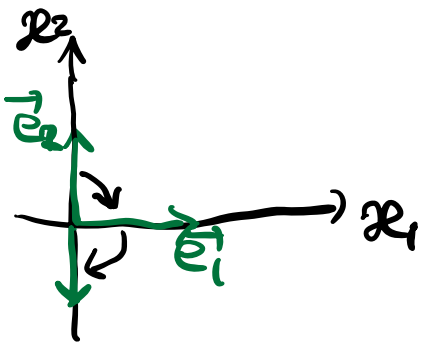
Example of non-diagonalizable matrix:

$B = \begin{bmatrix} 1 & -2 \\ 2 & 5 \end{bmatrix}$ Eigenvalue $\lambda = 3$

Eigenvector $\vec{v} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

Example

$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$



Rotation by 90° clockwise

There **cannot** be any $\lambda \in \mathbb{R}$, $\vec{x} \in \mathbb{R}$ s.t. $A\vec{x} = \lambda\vec{x}$

- How about **complex** eigenvalues?

Complex Numbers

complex number $z = a + bi$

↑ real part, ↑ complex part, $a, b \in \mathbb{R}$

- $i^2 = -1$
- $(a + bi) + (c + di) = (a + c) + (b + d)i$
- $(a + bi) \cdot (c + di) = (a \cdot c - b \cdot d) + (ad + bc)i$
- $\bar{z} = a - bi$ (complex conjugate)

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} -i \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ i \end{bmatrix} = i \begin{bmatrix} -i \\ 1 \end{bmatrix}$$

complex eigenvectors — complex eigenvalues

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} i \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -i \end{bmatrix} = (-i) \begin{bmatrix} i \\ 1 \end{bmatrix}$$

Important Message #2	A characteristic polynomial derived from an $n \times n$ matrix always has n roots (solutions), some possibly complex

$A\vec{x} = \lambda\vec{x}, \quad \vec{x} \in \mathbb{C}^n, \ \lambda \in \mathbb{C}$

Example

$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \qquad \det(A - \lambda I) = 0$

$$\begin{vmatrix} -\lambda & 1 \\ -1 & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 + 1 = 0 \Rightarrow \lambda^2 = -1$$

$\lambda_1 = -i$, $\lambda_2 = +i$ } complex conjugates

$\lambda_1 = -i$:

$[A - \lambda_1 I]\vec{x} = \vec{0}$

$A - \lambda_1 I = \begin{bmatrix} i & 1 \\ -1 & i \end{bmatrix}$

$$\begin{bmatrix} i & 1 & 0 \\ -1 & i & 0 \end{bmatrix} \xrightarrow{\cdot i} \sim \begin{bmatrix} i & 1 & 0 \\ -i & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} i & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$i \cdot x_1 + x_2 = 0 \Rightarrow x_2 = -ix_1$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ -ix_1 \end{bmatrix} = \begin{bmatrix} 1 \\ -i \end{bmatrix} \cdot x_1 \qquad \vec{x} \in \text{Span}\left\{\begin{bmatrix} 1 \\ -i \end{bmatrix}\right\}$$

$$\text{is } \begin{bmatrix} i \\ 1 \end{bmatrix} \in \text{Span}\left\{\begin{bmatrix} 1 \\ -i \end{bmatrix}\right\} ?$$

Similarly for $\lambda_2 = i$, $\vec{x}_2 = \begin{bmatrix} -i \\ 1 \end{bmatrix}$

- If λ is complex eigenvalue then so is its complex conjugate $\bar{\lambda}$.
- If $\vec{v}$ is a complex eigenvector then so is $\bar{\vec{v}}$.

Example continued

check $D = P^{-1}AP$ holds with $D = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix}$

and $P = \begin{bmatrix} i & -i \\ 1 & 1 \end{bmatrix}$

$\Leftrightarrow PD = AP$